

Analysis of CEX-DEX Arbitrage Opportunities with Hidden Markov Models

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Abstract

Web3 technologies and decentralised finance (DeFi) have not only revolutionised finance, but they have also given rise to completely different concepts in terms of how we handle our wealth and how we utilise the Internet, which is still an unreliable medium for the exchange of information. To overcome trust issues, cryptographic tools were introduced and these days we even put our digital assets on the blockchain. These new technologies enabled us to make most financial solutions decentralised. A widely popular financial product that is implemented on blockchains is a constant product market maker (CPMM). A CPMM can be used to swap digital assets between mutually untrusting parties by relying on prices calculated from its bonding curve. A blockchain-based automated market maker (AMM) quotes prices that usually lag behind external market prices, and this price difference can be exploited by automated bots, aka. arbitrageurs. The question we investigate in this work focuses on this issue. More specifically, we experiment with different approaches from financial mathematics to account for stochastic price changes, perform extensive measurements to quantify the characteristics of the DeFi market, focusing on mainnet Ethereum and develop stochastic grid optimisers to fit model parameters to actual AMM market data that we extract from the logs of the Ethereum virtual machine. We report on model performance that is more aligned with reality compared to previous works that have relied on naive two-parameter Black-Scholes (BS) models. Our results reveal that hidden Markov models can give better estimates for both the number of arbitrage trades and the arbitrage profits when combined with a jump diffusion model and a two-parameter BS model.

CCS Concepts

• **Computing methodologies** → **Simulation evaluation**; *Learning in probabilistic graphical models*; • **Applied computing** → **Economics**.

Keywords

cryptocurrency, arbitrage, DEX, CEX, hidden Markov model, statistical arbitrage, blockchain, market microstructure

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1 Introduction

In the Web3 era, users of the Internet not only consume and produce content, but they also manage their digital assets with the aid of decentralised technologies. The ownership of these digital tokens can be transferred using cryptography (i.e., signing messages with secret keys) and the integrity of the databases storing the ownership of the assets can be ensured with blockchain technology, which is reliant on a decentralised network of computers keeping the distributed ledger synchronised. Among others, Ethereum [9, 10] with its fully fledged Turing-complete virtual machine (EVM) is capable not only of registering native ETH transfers but also of executing arbitrary program code (smart contracts) and this invention has given rise to the concept of decentralised exchanges [11, 31] (DEXs). DEXs, as opposed to their centralised counterparts (CEXs) entail no counterparty risk as the traded assets do not have to be deposited at a centralised entity. Owing to this (and other) favourable characteristics, DEXs have become increasingly popular among blockchain users. Considering the figures, the trading volume for actively traded pairs has shifted from CEXs to DEXs after their inception, and in January 2023, 60% of the total volume was attributable to on-chain DEX transactions [4]. The Ethereum blockchain orchestrates the trade of several billion dollars each day via its DEXs. As of 23 September 2025, the Uniswap [42] protocol boasts \$4.20B in total value locked (TVL), its version 2 (V2) [2] accounting for \$1.96B. The protocol has undergone several updates, with V4 being the latest, though older versions remain in use.

DEXs can only execute trades when a new block is added to the chain. Such a slow mechanism results in losses (see LVR in [35]) for the liquidity providers (LPs) when the quoted prices lag behind CEX prices. In other words, price differences are exploited by arbitrage traders who earn riskless profit [19] on each trade by buying from the automated market maker (AMM) when the price is low and selling the tokens on a CEX for a higher price. This obviously also works in the opposite direction. Such arbitrage losses are well known from traditional finance contexts [21] and approaches to mitigate this issue are actively being researched [13]. On the other hand, AMMs charge a fee for each swap (Uniswap V2 for example charges a flat fee of 0.3%) and fee revenues should balance arbitrage losses for LPs to make AMMs an attractive financial product to



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investors. The question before us is, what is the optimal value of this fee? This issue is also being actively investigated [20]. LPs make money on both noise and arbitrage trades by collecting exchange commission, therefore to find out whether it was profitable to invest in an AMM, one should consider the fees collected. Additionally, the LP position's value at t_0 should be compared to the position's value when it is being liquidated. Our empirical data will shed light on this aspect.

A significant proportion of the CEX-DEX arbitrage profit appears as maximal extractable value (MEV) [18]. Some try to find an answer to the question whether MEV is harmful to the ecosystem [39, 44]. Note that while this work explores only CEX-DEX arbitrage opportunities, there are other forms of arbitrage trades, see for example [24], [6] and [28]. The fact that one can only generate a high MEV block with sophisticated algorithms leads to the separation of roles of block proposers and builders [23], and Ethereum currently employs a proposer-builder-separation (PBS) [12] protocol connecting the highly competitive block building process [3] with its consensus layer. As mentioned above, MEV affects AMMs as well, primarily in the form of losses incurred by arbitrage trades. This has been empirically quantified in [22] and we take a somewhat similar **data-driven** approach in this work, albeit with a dataset that is augmented by an indexer. More specifically, a software architecture was designed to collect data from several sources. Firstly, an Ethereum consensus client and an execution layer node are synchronised to log swap events on Uniswap V2 AMMs. Then we also connect to the spot price API provided by Binance¹ to generate timeseries for the ETH/USDT exchange rate. Finally, we retrieve PBS bids from the Flashbots² relay that were placed on the blocks under consideration. With this architecture, we perform fine-grained measurements over several days with 1s granularity. Then we calculate the daily drift and volatility of logarithmic returns of a handful of cryptocurrency pairs and demonstrate that the market can change very rapidly, even over a short period of time. Then the same dataset is analysed to extract stochastic jumps and to determine their average sizes for the pairs under consideration. This was performed to underscore our viewpoint that jumps [32] in the stochastic price evolution models are indeed necessary to calculate CEX-DEX arbitrage figures accurately. Then we dissect our DEX dataset and merge it with the CEX dataset to count the number of arbitrage swaps and the amount of arbitrage profits for each day. These serve as target values in the **stochastic grid optimiser** that tries to find the optimal values for the stochastic parameters to reproduce the real-life data (obviously only in some average sense). Our findings show that it is fairly easy to obtain the same numbers for arbitrage metrics when the market is in a dormant state (i.e. low volatility and drift) with a naive two-parameter model; however, it becomes challenging when crashes/rallies occur. To overcome this difficulty, the two-parameter stochastic model is extended and a Monte Carlo simulator is designed for the simulation of the extended model. The related source code and the dataset is publicly available and our results are reproducible using a public Zenodo repository³.

¹<https://www.binance.com/en/binance-api>

²<https://www.flashbots.net/>

³<https://doi.org/10.5281/zenodo.1834784>

The expected amount of arbitrage trades, collected fees and trade volume are approximated by introducing a stochastic model for the spot price. The precision of these approximations depends heavily on how much the introduced mathematical model overlaps with reality. Our first try is a geometric Brownian motion (GBM) model. GBM has been the workhorse of financial mathematics since the seminal work of Black-Scholes [7], but since then more sophisticated models have been developed. The Ornstein-Uhlenbeck stochastic differential equation (SDE) [41] is also frequently employed to describe mean-reverting effects [30]. Merton's jump-diffusion model (MJD) [32] is used when stochastic jumps should be treated on a different footing. In this work, we attempt to devise a regime-switching hidden Markov model (HMM) [40], whereby we employ two SDEs, a GBM and an MJD and the states are determined by a 2-state hidden Markov chain.

In addition to the stochastic optimiser we present several interesting results that we obtained by analysing our CEX, DEX and PBS datasets. The figures we present below indicate that there is a strong correspondence between the change in CEX prices (volatility), arbitrage profits and PBS bids (MEV). To the best of our knowledge, this is the first endeavour to merge these datasets and to investigate the effect of the change in CEX prices on PBS and Uniswap V2 AMMs, which is a clear indication of the presence of CEX-DEX arbitrageurs in the ecosystem.

The contributions of this work are as follows:

- A blockchain indexer architecture is introduced.
- Arbitrage profits are analysed using empirical measurements and exchange rates are subjected to timeseries analysis.
- Regime-switching hidden Markov model was introduced to the stochastic price process.
- A stochastic Monte Carlo optimiser is devised and its results are evaluated to fit model parameters to reproduce real-life data.

This work is structured as follows. After having discussed the related work in Sec. 2 the mathematical model along with the employed SDEs are introduced in Sec. 3. Then in Sec. 4 our data collection methodology is discussed. Our heuristics approach to CEX-DEX arbitrage is presented in Sec. 5. The results of the Monte Carlo optimisation problem are discussed in Sec. 6, then we conclude our work.

2 Related Works

Although classical AMMs had been in existence for many years the seminal works of A. Capponi et al. [15] and J. Milionis et al. [35], [33] set the premise for blockchain-centred AMM research. Extensions have also appeared [36], [29]. Inefficiencies in AMMs [8, 37] were pointed out by several scholars of the research community, refer to [5] and [16, 17, 43, 46]. For a comprehensive guide on AMMs the reader is referred to the SoK in [45]. Numerical approaches with Ornstein-Uhlenbeck [41] processes are presented in [30]. The authors in [38] follow a path that is somewhat similar to our endeavour in the sense that they also analyse the spot price of ETH and explore arbitrage opportunities between exchanges.

3 Mathematical model

For a detailed explanation on how decentralised exchanges work and what kind of transaction fees one has to pay when using an on-chain AMM refer to [1] and to [25].

When analysing the exchange rate data of the USDT/ETH pair we found that the underlying stochastic process that drives the price is definitely not a two-parameter one (e.g. GBM). A two-parameter SDE (with drift and volatility) is capable of describing the market near its equilibrium when neither crashes nor rallies take place. Cryptocurrency markets, however, are different. Price jumps exceeding the instantaneous volatility are frequently observed and an accurate stochastic model should take into account these changes on a different footing. Merton's jump-diffusion model is a prominent example among these models that can be utilised in this case. To ease comprehension, let us repeat these well-known SDEs:

Geometric Brownian Motion (GBM):

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where S_t is the asset price, μ the drift, σ the volatility, and W_t a standard Brownian motion [7]. This SDE models continuous asset price evolution with constant drift and volatility. This model was employed in many AMM-related works such as [26, 33, 35].

Merton Jump Diffusion Model (MJD):

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_t^- dJ_t,$$

where $J_t = \sum_{i=1}^{N_t} (Y_i - 1)$ with N_t a Poisson process of intensity λ and Y_i i.i.d. jump sizes [32]. This model extends GBM by incorporating sudden jumps in asset prices driven by a Poisson process. Notwithstanding the fact that the MJD is a slightly more sophisticated model than the GBM, it is still a primitive model in the sense that it assumes that prices can "only" drift, fluctuate and jump. However, when a crash occurs or when a rally takes place the market gets overheated and it, so to speak, switches itself into a completely different state. Such behaviour can be described by a hidden Markov model (HMM), whereby one defines a finite Markov chain with two possible states and lets the price evolve according to an alternating GBM and an MJD SDE as enumerated below:

Regime-Switching Model with Markov Chain:

Let $\{X_t\}_{t \geq 0}$ be a continuous-time Markov chain on the state space $\{1, 2\}$ with its state transition matrix:

$$Q = \begin{pmatrix} q_{11} & q_{12} \\ q_{21} & q_{22} \end{pmatrix},$$

where q_{ij} denotes the transition intensity from state i to j . The identities $q_{11} + q_{21} = 1$ and $q_{12} + q_{22} = 1$ are assumed.

Using the same notations as above, the spot price S_t then evolves as

$$dS_t = \begin{cases} \mu_{\text{GBM}} S_t dt + \sigma_{\text{GBM}} S_t dW_t, & \text{if } X_t = 1, \\ \mu_{\text{MJD}} S_t dt + \sigma_{\text{MJD}} S_t dW_t + S_t^- dJ_t, & \text{if } X_t = 2, \end{cases}$$

where W_t is a standard Brownian motion and $J_t = \sum_{i=1}^{N_t} (Y_i - 1)$ with N_t a Poisson process of intensity λ and Y_i i.i.d. jump sizes. The time until the first jump T is assumed to follow an exponential distribution:

$$P(T > t) = e^{-\lambda t}.$$

This model allows the asset price dynamics to switch randomly between a pure diffusion (GBM) and a jump-diffusion (Merton) according to an underlying Markov chain.

The main focus of this work is the misprice process between a CPMM and the external market price, defined as the natural logarithm of the spot price and the price quoted by the AMM:

$$z_t = \ln \left(\frac{P_{\text{CEX}}^t}{P_{\text{DEX}}^t} \right), \quad \forall t > 0,$$

and usually it is assumed that whenever P_{CEX} is Markovian and the arbitrageurs arrive according to a Poisson process, z_t is also a Markov chain (if a discrete approximation is taken) whose stationary distribution exists under mild conditions regarding the price process itself, for details see [33]. The number of blocks per day containing arbitrage trades can be calculated from this stationary density by integrating it from $-\infty$ to $-\gamma$ and from γ to ∞ . If a two-parameter GBM process is assumed with symmetric parameters (i.e. $\mu = \frac{1}{2}\sigma^2$) and the daily volatility is 5% then 12.3% of the blocks should contain an arbitrage trade with 30bp fees and 12s blocktime (Ethereum), see [33] for more numerical details of this analysis. Also note that the stochastic model in [33] neglects the presence of CEX exchange fees. As a consequence, such a model is assumed to give much higher estimates for the number of trades and arbitrage profits.

4 Data Collection

The collection of cryptocurrency market data is relatively simple these days as major exchanges (Binance, Kucoin, Gateio, etc.) provide a spot price API for free. In our case, exchange rates with 1s resolution are collected from a webstream API provided by Binance and they are then stored in a MySQL database. On the other hand, DEX rates are less straightforward to collect. While there are some data providers such as Dune, collecting a large amount of data can be costly and time-consuming. The most universal way to perform this is to have a synchronised blockchain node and collect the data from log events. To that end, we have chosen to rewrite the open-source implementation of the Ethereum protocol in geth⁴. Once an event is generated we log everything in the MySQL database, implying that there is no need for archive sync mode. Our node runs in snap sync mode to save computational resources and bandwidth. We found that this method is sufficient for indexing and our database backups grow by merely a few megabytes per day. In order to offload computational tasks from the node, our geth client writes everything to an HTTP socket, which is then read by a client software developed in Rust. The Rust code is then responsible for post-processing the data and for creating the MySQL backups. The distinct division of tasks between software components enables one to efficiently index the blockchain. To understand how these software components are connected please refer to Fig 1.

While the Internet abounds with tutorials relying on JavaScript-based libraries (web3) to index the Ethereum blockchain, we found that they are not fast enough and such an implementation frequently misses swap events, rendering the entire measurement process meaningless. This problem becomes even more pronounced when volatility is large and the market fluctuates heavily, as the

⁴<https://geth.ethereum.org/>

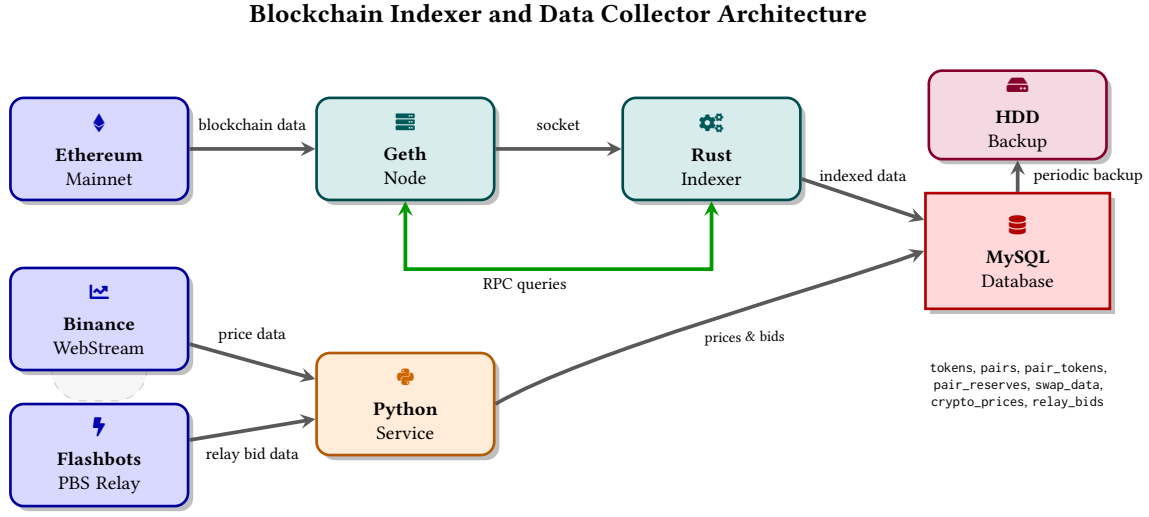


Figure 1: Overview of the Ethereum blockchain indexer architecture showing data flow from external sources through processing components to the database storage layer.

number of AMM trades can grow severely and such a naive approach is unable to keep track of the incoming transactions. We save chain-related data in the database when

- (1) a transaction occurs on an AMM,
- (2) a new block arrives.

For the first case, the transaction hash, the pair address, the amounts in and out and a timestamp are recorded. When the second event occurs, we query the node to extract `reserve0` (X) and `reserve1` (Y) values using a standard Uniswap V2 AMM ABI. The transaction hash can be used to inspect individual transactions manually and to gain more insight into transaction-level data.

4.1 Timeseries Analysis

For some of the most popular cryptocurrencies (ETH, BTC, AAVE, CHZ, CVR, LINK, NMR and UNI), we calculated the daily mean return (μ) and volatility (σ), which allowed us to highlight the heterogeneous risk–return characteristics of different cryptocurrencies and the presence of volatility clustering across time. For the numerical results, refer to Table 1. In addition, we examined cryptocurrency–USDC trading pairs using high-frequency data collected at one-second intervals from the Binance WebStream API. Our results are presented in Table 2. Extreme price movements, referred to as jumps, were defined as returns exceeding 9.5 standard deviations of the respective asset’s return distribution. For each trading day, we measured both the frequency of these jumps and their average size expressed as percentage returns.

5 CEX-DEX Arbitrage

Empirical evidence shows that the instantaneous price quoted by a CPMM (i.e. $P_{\text{DEX}} = \frac{Y}{X}$ with Y denoting the amount of asset y and X the amount of asset x held by the AMM) closely follows the external market price. This is due to the fact that whenever P_{DEX} deviates from the P_{CEX} it becomes profitable to execute a trade against the AMM and immediately sell the acquired asset on a CEX. Note that

in [34] this is referred to as price discovery in Proposition 2.1. The extent of deviation that is necessary for an arbitrage trade to take place depends on many factors, such as CEX and DEX fee levels and time to execute trades, to name a few. Every Uniswap V2 charges a flat fee of 30 basis points or 0.3%. CEX fee levels, on the other hand, can change over time and not every exchange charges the same amount of commission; therefore this part of the analysis is taken as a black box. To simplify our discussion related to the empirical data, we assume that an arbitrage trade takes place when the price quoted by the AMM is less favourable than P_{CEX} , in other words, the arbitrageur buys from the AMM when P_{DEX} is lower than P_{CEX} and she sells to the AMM when the opposite is true. We use this heuristic, which is similar to the window method described in [14], to identify arbitrage trades against the AMM. More specifically, data was collected using the aforementioned methodology and AMM trades were specified according to whether it was profitable to execute the two legs of the trade, one on a CEX and the other on a DEX. Note that in some cases, false positives could have occurred because, in some cases, even though it was financially rational to execute the trade, the agent might not have executed the CEX leg, and as we had no means to trace CEX transactions, there is no telling whether the CEX leg was executed or not. Simply we just assumed that every time there was an arbitrage opportunity between a CEX and the DEX, it was exploited. If we assume that market participants are rational, this is indeed a proper heuristic. To give an idea of how the market evolved during our measurement period, we refer the reader to Fig. 2. In this figure, we plot the USDT/ETH spot price, P_{DEX} for one of the largest Uniswap V2 USDT/ETH AMM (see the figure for its address) and the corresponding misprice process (z_t). Fee levels are indicated by two horizontal lines at -0.003 and $+0.003$. The red vertical lines indicate the points in time when a trade against the AMM occurred and when the acquired tokens could have been sold at a CEX for a better price, thereby obtaining a riskless profit. The first reassuring thing we see in this figure is that arbitrage trades take place when the misprice

Date	ETH		BTC		AAVE		CHZ		CRV		LINK		NMR		UNI	
	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
09-18	1.45	1.81	0.19	0.95	2.77	2.89	3.24	3.33	6.38	5.46	3.73	4.57	3.45	6.00	1.90	2.93
09-19	-2.72	1.77	-1.56	0.85	-3.62	3.14	-4.28	3.43	-3.90	3.92	-4.32	4.75	-2.43	5.74	-4.93	3.29
09-20	0.42	0.95	0.28	0.39	-0.33	1.92	1.25	2.27	1.94	3.06	-0.68	3.12	-1.26	3.38	0.57	1.82
09-21	-0.42	0.98	-0.34	0.38	-1.08	1.86	-0.99	2.15	-2.06	2.84	-1.42	2.71	-4.13	3.65	-1.41	1.76
09-22	-6.29	3.61	-2.12	1.54	-9.19	7.30	-7.46	5.04	-12.29	6.72	-6.54	6.11	-11.67	8.04	-10.14	6.75
09-23	-0.38	1.61	-0.77	0.77	3.18	4.30	1.65	3.75	-3.15	3.14	0.65	3.54	11.91	12.76	-1.15	2.40
09-24	-0.90	2.04	1.07	0.98	-0.50	3.28	0.23	3.05	3.52	4.25	-0.51	4.12	-2.62	6.92	-1.96	3.10
09-25	-5.99	4.01	-3.54	1.54	-5.30	4.85	-3.52	4.28	-8.23	6.24	-5.71	5.91	-7.62	9.23	-5.86	4.89
09-26	3.01	3.12	0.08	1.42	3.15	4.14	2.64	3.95	3.58	5.06	3.04	5.62	10.05	9.80	1.62	4.46
09-27	-0.23	1.42	-0.00	0.55	-1.25	1.74	-1.17	2.07	-0.06	2.57	-0.43	2.98	-3.43	6.85	-0.18	2.35

Table 1: Daily mean return (μ) and volatility (σ) values (in %) for cryptocurrency trading pairs against USDC. Data collected via Binance WebStream API with 1-second resolution. Notable variations in both μ and σ values across different cryptocurrencies and trading days highlight the heterogeneous risk-return profiles and temporal volatility clustering in digital asset markets.

Date	ETH		BTC		AAVE		CHZ		CRV		LINK		NMR		SNX		UNI	
	C	S (%)	C	S (%)	C	S (%)	C	S (%)	C	S (%)	C	S (%)	C	S (%)	C	S (%)	C	S (%)
09-18	11	0.11	12	0.05	11	0.16	11	0.34	72	0.28	0	0.00	27	0.71	0	0.00	7	0.23
09-19	44	0.11	18	0.07	14	0.17	12	0.46	20	0.22	5	0.22	24	0.73	0	0.00	27	0.24
09-20	5	0.11	2	0.04	5	0.17	15	0.31	14	0.21	4	0.26	7	0.70	0	0.00	2	0.29
09-21	7	0.16	4	0.06	0	0.00	8	0.40	8	0.26	3	0.24	7	0.75	0	0.00	5	0.18
09-22	279	0.13	186	0.06	189	0.28	94	0.36	212	0.26	92	0.23	48	0.87	0	0.00	235	0.26
09-23	14	0.11	21	0.05	7	0.66	26	0.43	26	0.22	3	0.55	27	1.21	0	0.00	5	0.23
09-24	51	0.11	40	0.05	26	0.20	39	0.32	45	0.22	12	0.22	41	0.78	0	0.00	25	0.20
09-25	283	0.12	148	0.05	105	0.18	70	0.33	162	0.23	30	0.23	54	0.96	0	0.00	123	0.19
09-26	130	0.11	136	0.05	52	0.17	50	0.37	92	0.22	15	0.22	82	0.85	0	0.00	66	0.20
09-27	32	0.11	10	0.05	3	0.15	13	0.32	9	0.24	0	0.00	40	0.78	0	0.00	16	0.26

Table 2: Daily jump count and average jump size (in %) for cryptocurrency trading pairs against USDC. Jumps are defined as returns exceeding 9.5 standard deviations of the respective symbol's return distribution. Count represents the number of jumps per day, while Size shows the average magnitude of jumps as percentage returns.

Date	Swaps	ARB	ARB%	Noise	ETH Vol	ARB Vol (\$10 ⁶)	Noise Vol (\$10 ⁶)	Fees (\$)	ARB Profit (\$)	Vol/Profit	μ (%)	σ (%)
09-18	2199	259	3.6	1940	604.1	1.1	1.7	8327.8	944.5	2939.1	1.45	1.81
09-19	1893	231	3.2	1662	383.9	0.5	1.3	5210.3	232.7	7464.4	-2.72	1.77
09-20	1611	148	2.1	1463	231.7	0.3	0.7	3115.7	225.3	4609.6	0.42	0.95
09-21	1614	203	2.8	1411	174.7	0.3	0.5	2344.1	166.6	4690.2	-0.42	0.98
09-22	2061	342	4.8	1719	1710.0	3.8	3.4	21558.2	22341.5	321.6	-6.29	3.61
09-23	1979	321	4.5	1658	750.7	0.9	2.2	9406.5	1020.5	3072.5	-0.38	1.61
09-24	1816	188	2.6	1628	615.3	1.1	1.5	7671.7	827.7	3089.7	-0.90	2.04
09-25	2877	385	5.3	2492	1624.1	3.2	3.3	19402.7	3756.8	1721.5	-5.99	4.01
09-26	2402	364	5.1	2038	1243.6	2.1	2.8	14698.3	3172.5	1544.3	3.01	3.12
09-27	3084	286	4.0	2798	675.4	0.5	2.3	8129.8	601.2	4507.6	-0.23	1.42
Average	2153.6	272.7	3.8	1880.9	801.4	1.4	2.0	9986.5	3328.9	3396.1	-1.21	2.13

Table 3: Daily arbitrage analysis for 0x0d4a11d5EEaC28EC3F61d100daF4d40471f1852 with GBM parameters using USD/ETH prices from Binance (2025-09-18 to 2025-09-27). Shows swap activity, arbitrage opportunities, volumes, fees, profit, volume to profit ratio, daily drift (μ) and volatility (σ) in percentage. TVL $\approx$ 20.368 ETH.

process wanders outside the $[-\gamma, \gamma]$ band. These empirical results underpin the theoretical findings of [33]. On the other hand, note that there is uncertainty as to what the external market prices are and how arbitrage profits should be calculated. In this work, P_{CEX} is the market price exactly when the block is produced; however in

reality, one can use any market price after the block's timestamp, but this uncertainty was not investigated by us.

Moving on, we performed a vivisection on this data with a daily resolution to find out how many swaps were executed against this AMM, among these, how many of them were arbitrage trades (according to the aforementioned heuristics). Using these figures

we further calculated the arbitrage trade volume, the amount of fees collected and the total arbitrage profit per day. Our measurement results are compiled into Table 3. The results of additional AMMs are delegated to the Appendix. The column denoted as ARB% indicates the amount of blocks containing arbitrage swaps divided by the number of blocks per day (7200). Any realistic stochastic models should reproduce these values when the stationary distribution is integrated from $-\infty$ to $-\gamma$ and γ and ∞ . In [27] the authors state that *over 96.5% of the blocks in our sample do not have arbitrage opportunities*. Compare that to the average of the ARB% column of Table 3 to find a reassuring match of numbers (our result is 3.8% on average).

Finally, we used our PBS datasource to collect bids and create categories for both the number of swaps and the maximal bid amount for each day. These results are presented in Table 4. Observe that when the volatility (and the absolute value of the drift) is large, the amount of high-value USDT swaps and high-MEV blocks increases, indicating the presence of automated CEX-DEX arbitrage bots executing massive trades on the AMMs and placing high-value bids on the PBS relays.

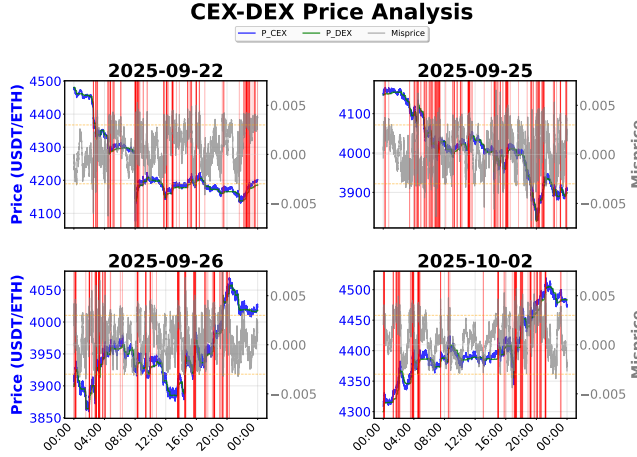


Figure 2: CEX, DEX price processes on a USDT/ETH Uniswap V2 AMM with the respective misprice process for certain days of our measurement period. Red horizontal lines indicate profitable DEX swaps, whereby the arbitrageur swaps tokens in such a direction that yields profit, should the tokens be sold on a CEX.

Besides the AMM with the largest TVL, we analysed 7 additional AMMs (see the Appendix for their address) evaluating the same metrics as above (daily swap count, daily total profit, daily accumulated fees and volume to profit ratio). Our results are presented in Fig 3. Oddly enough, all of these AMMs exhibit very similar behaviour when these metrics are considered, presumably due to the fact that a significant proportion of the trade is generated by automated bots who execute trades according to the changing market conditions and they seem not to differentiate between AMMs. Whenever the market changes they execute a trade against the AMM, should the trade yield an immediate riskless profit. The daily swap count is proportional to the TVL of the pool, which is attributable to the

fact that pools with larger TVL can quote better prices (i.e. less slippage) and these pools are more frequently utilised by traders. The daily arbitrage profits and the fees collected are proportional to the TVL, as expected. Another interesting metric is the volume-to-profit (VtP) ratio. It is often conjectured that the VtP ratio is constant, however we have seen no theoretical evidence of this. We plot this metric for each AMM in the bottom right subplot of Fig. 3.

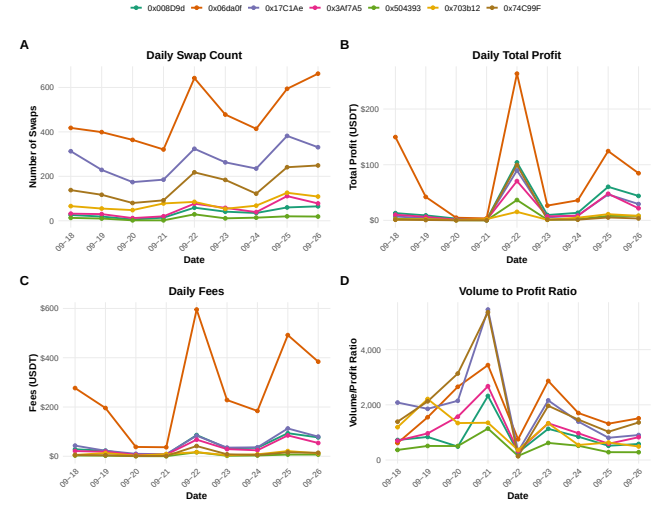


Figure 3: Daily arbitrage performance metrics across seven ETH/USDT trading pairs. Panel analysis shows: (A) transaction volume, (B) realised profits in USDT, (C) trading fees generated, and (D) volume-to-profit efficiency ratios. Data represents algorithmic arbitrage opportunities identified between decentralized exchange pairs.

6 Stochastic Optimisation

Next, we turn our attention to the fine-tuning of the model parameters. More specifically speaking, we make attempts to fit the SDE model parameters such that the resulting price and AMM dynamics resemble real-life trading data. As we made extensive numerical experiments and collected both CEX and DEX data over a couple of thousands of blocks as discussed above, we are in a position to calibrate the GBM, MJD and HMM parameters to obtain results with synthetic data that are similar to the one measured in real-life. The optimisation problem we are about to introduce below shall vary the following model parameters (in accordance with the notation above):

- (1) GBM: σ_{GBM} , μ_{GBM} ,
- (2) MJD: σ_{MJD} , μ_{MJD} , λ_{MJD} , Y_i and
- (3) HMM: q_{11} and q_{22} .

Fee levels are set to 30bp (symmetric) to simulate classical Uniswap V2 AMMs. Block times are set to 12s (a typical value in Ethereum). Another two variables we had to optimise for are $\eta_S = \frac{\sigma_{MJD}}{\sigma_{GBM}}$ and $\eta_D = \frac{\mu_{MJD}}{\mu_{GBM}}$, in other words the ratio between the two parameters of the GBM and the MJD model. We found that for most of the cases, the HMM works better with $\eta_S > 1$ and $\eta_D \leq 1$.

Table 4: Daily swap transaction counts with MEV activity (measured as the distribution of the maximal PBS bid, captured on the Flashbots relay for each category) for the AMM at 0x0d4a11d5EEaaC28EC3F61d100daF4d40471f1852. The stochastic parameters are given in %. The sum of the maximum of bids is given in ETH in the last column.

Date	μ	σ	Swap Counts by USDT Amount						Max	MEV Bids by ETH Value					Total
			[0-10]	[11-50]	[51-500]	[500-10K]	[10K-50K]	[50K+]		0-0.1	0.11-0.5	0.51-0.99	1-5	5+	
09-18	1.45	1.81	335	388	845	566	64	1	67.4K	6604	176	15	4	0	178.76
09-19	-2.71	1.77	419	436	615	385	38	0	44.4K	6572	213	16	2	0	159.85
09-20	0.42	0.95	444	484	390	279	12	2	51.0K	6503	241	20	3	2	173.53
09-21	-0.42	0.98	469	366	507	261	11	0	49.8K	6534	209	11	2	0	139.34
09-22	-6.29	3.61	346	400	640	554	99	22	784.2K	6413	292	28	20	13	402.23
09-23	-0.38	1.61	412	400	572	510	84	1	90.0K	6579	173	5	3	2	178.45
09-24	-0.90	2.04	452	393	536	373	58	4	110.4K	6498	178	18	15	3	204.40
09-25	-5.99	4.01	385	637	1014	646	189	6	162.7K	6486	233	30	12	2	247.04
09-26	2.99	3.12	455	585	644	598	112	8	214.3K	6617	169	8	5	2	189.67
09-27	-0.23	1.42	784	607	1063	573	54	3	153.3K	6624	124	16	11	0	145.60

6.1 Monte Carlo Optimisation

To reproduce real-life swap counts and arbitrage profits, we generate a discrete series of spot price data using the GBM model and simulate trades against our hypothetical AMM. The commands to run our simulation framework can be found in the repository (see the bottom of page 2. for URL).

For the sake of simplicity we simulate a 24h trading window of 7.200 blocks, create an AMM object with initial liquidity $X = 10.081$ ETH and $Y = 45.724.211$ USDT (this roughly equals the TVL of the Uniswap V2 ETH-USDT AMM we analysed above). X and Y do not seem to change significantly over a few-day period, and we opted to keep these values constant (besides the executed trades that of course change the AMM's liquidity). The average number of trades and arbitrage profits in USDT for each day are presented in Table 5. The arbitrage profit is calculated as the product of the amount of ETH exchanged and the difference between P_{CEX} and P_{DEX} . While the figures demonstrate the proportionality between volatility and the number of arbitrage opportunities, it is clear that the GBM model substantially underestimates these values. The error is the most pronounced for September 22nd, when the measured arbitrage profits was in the region of 22K USDT, the GBM model resulted in merely 591 USDT of profits.

Date	μ	σ	Avg Trades	Avg Profit
09-18	0.0145	0.0181	255.73	74.86
09-19	-0.0272	0.0177	351.13	98.21
09-20	0.0042	0.0095	116.70	9.06
09-21	-0.0042	0.0098	114.96	9.59
09-22	-0.0629	0.0361	508.46	591.22
09-23	-0.0038	0.0161	199.76	43.96
09-24	-0.0090	0.0204	238.10	85.14
09-25	-0.0599	0.0401	515.56	740.88
09-26	0.0301	0.0312	371.56	324.03
09-27	-0.0023	0.0142	151.76	26.92

Table 5: Average simulation results (number of trades and arbitrage profits in USDT) over 30 runs for different GBM parameter (μ and σ) values.

To correct these inaccuracies, we use the GBM and the MJD SDEs and simulate the combination of these two using an HMM, switching states according to the probabilities q_{11} , q_{12} , q_{21} and q_{22} . The simulator is evaluated along the same set of metrics and the results are compiled into Table 6.

For every discrete timestep, an increment for the spot price is calculated using either the GBM or the MJD SDE. This choice is made according to the probabilities q_{11} , q_{12} , q_{21} and q_{22} to determine whether the market is in a dormant state with no jumps or a crash/rally is taking place. For each combination of parameters, we run several Monte Carlo simulations ($N_{\text{runs}} = 20$) and calculate the number of trades and the arbitrage profit in USDT. Then these values are averaged and a loss value is calculated as follows:

$$\text{loss} = \frac{|\bar{T} - T^*|}{T^*} + \frac{|\bar{P} - P^*|}{P^*},$$

where $\bar{T}$ and $\bar{P}$ denote the average number of trades and profits, while T^* and P^* are the target values. The results of the MC grid optimiser are presented in Table 6. Regarding these results, we make the following observations. When μ and σ are relatively small, the number of profitable trades and the amount of arbitrage profits are also small and the stochastic model can easily reproduce these value by keeping q_{11} near 0.99, implying that the HMM is almost always in the GBM state. According to our experience, once a suitable set of parameters were found, it was possible to fine-tune the final results by using different η_D and η_S values. While at each point a significant amount of loss remains, the target variables are near the measured ones. Note that the primary goal of this work was to show that an HMM can be more flexible than a naive GBM approach, rather than reproducing the measured values with high precision. This could have been almost impossible as with a certain set of parameters we could measure only one trajectory. To measure more trajectories, one would have to wait until the SDE parameters appear again and average the obtained measurement values. This takes an infeasible amount of time, therefore we decided not to pursue this direction.

The most interesting data point in our dataset is 2025-09-22 when a substantial amount of arbitrage profit (approximately \$22300) was generated on the AMM we analysed. To reproduce these figures, we see that the MC grid optimiser found a parameter combination of $q_{11} = 0.8583$ and $q_{22} = 0.2333$. This clearly indicates the power

Table 6: Results of the MC grid optimiser with various simulation parameters. The tool finds the best possible combination of stochastic parameters (HMM transition probabilities (q_{11} and q_{22}) the intensity (λ) and size of jumps ($|Y_i|$), drift (μ) and volatility (σ), η_D and η_S) that reproduce the number of arbitrage transactions and profit for each day we performed a measurement.

Date	q_{11}	q_{22}	λ	$ Y_i $	μ	σ	η_D	η_S	Target Tr	Final Tr	Target Pr	Final Pr	Loss
2025-09-18	0.9900	0.7667	0.000100	0.015083	0.0145	0.0181	2.00	0.50	258	233.5	944.5	1163.5	3.0%
2025-09-19	0.5950	0.1000	0.006700	0.000100	-0.0272	0.0177	5.00	2.50	231	322.8	232.7	240.9	31.6%
2025-09-20	0.9900	0.2333	0.000100	0.015083	0.0042	0.0095	1.50	0.50	148	119.1	225.3	364.8	31.4%
2025-09-21	0.9900	0.2333	0.000100	0.015083	-0.0042	0.0098	3.00	0.50	203	133.2	166.6	370.2	58.6%
2025-09-22	0.8583	0.1000	0.000100	0.030067	-0.0629	0.0361	11.00	0.50	342	465.8	22311.1	19130.7	36.1%
2025-09-23	0.9900	0.6333	0.000100	0.015083	-0.0038	0.0161	12.00	0.70	321	197.2	1020.8	936.1	51.8%
2025-09-24	0.9900	0.6333	0.000100	0.015083	-0.0090	0.0204	3.00	0.50	188	247.0	827.7	888.3	23.2%
2025-09-25	0.9900	0.9000	0.000100	0.015083	-0.0599	0.0401	2.00	1.00	385	518.2	3756.5	3470.5	49.0%
2025-09-26	0.9900	0.6333	0.000100	0.030067	0.0301	0.0312	6.00	1.00	363	368.3	3172.4	3889.1	16.0%
2025-09-27	0.9900	0.5000	0.000100	0.015083	-0.0023	0.0142	2.70	0.50	286	171.3	601.0	709.4	47.9%

of the HMM investigated by us in the present work. The transition probabilities tell us that in order to reproduce large arbitrage profits with few trades, the SDE must contain a jump part, but it should be turned on according to another stochastic process (HMM). This gives the MC grid optimiser the flexibility to find the best set of parameters in a larger space. On the downside, even our HMM simulator struggles with data points where the arbitrage profits are low (e.g. 2025-09-21). Corrections will be considered in upcoming works.

7 Conclusions

This work introduces a blockchain indexer architecture and discusses market dynamics between the external cryptocurrency markets and an on-chain CPMM. Our research laid its focus on the question of whether a two-parameter SDE can reproduce real-life figures of a Uniswap V2 AMM. The problem at hand were analysed using both empirical approaches and theoretical methods. The primary goal of this research work was to draw the theoretical model as close to reality as possible. Our endeavour commenced from the observation that cryptocurrency markets can be erratic and in many cases the daily volatility changes significantly over a short period of time. Such observations were also made when probabilists analysed returns on stocks and found that there is a need to introduce time-dependent volatility approaches. Here, a similar attempt was made, albeit with the help of a simple Markov chain that was able to describe the hidden Markovian nature of the cryptocurrency market. We collected data from three independent data sources and demonstrated that the change in market volatility has a direct bearing on the amount of arbitrage trades and the bids that are placed on PBS relays for a block to be then included in the canonical chain. With the help of our empirical data we were able to present evidence of CEX-DEX arbitrage trades that happen in the wild.

Regarding the limitations of this work, the fact that only a single DEX price source was analysed might slightly distort the results in terms of arbitrage profits. This discrepancy will be corrected in our upcoming works. When the arbitrage profits were calculated, we neglected the CEX fees, and this should also be investigated to deliver more accurate results. AMMs with concentrated liquidity [26] have not been analysed, Further endeavours in this direction will be

reported. Armed with the findings of this work, a natural extension of our results would be a stochastic model that can be utilised to calculate the optimal value of the exchange commission charged by the AMM which would lead to the largest possible trading volume (both noise and arbitrage trades).

8 Disclosures

The authors declare that they do not have any competing interests. Notwithstanding, the ideas and opinions expressed herein are those of the authors, rather than of any of these companies or their affiliates, and should not be construed as or relied upon in any manner as investment, legal, tax, or other advice.

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References

- [1] Austin Adams, Benjamin Y Chan, Sarit Markovich, and Xin Wan. 2023. *The Costs of Swapping on the Uniswap Protocol*. Papers 2309.13648. arXiv.org.
- [2] Hayden Adams, Noah Zinsmeister, and Dan Robinson. 2020. Uniswap v2 Core. <https://uniswap.org/whitepaper.pdf>. Accessed: 2025-10-01.
- [3] Maryam Bahrani, Pranav Garimidi, and Tim Roughgarden. 2025. Centralization in Block-Building and Proposer-Builder Separation. In *Financial Cryptography and Data Security*, Jeremy Clark and Elaine Shi (Eds.). Springer Nature Switzerland, Cham, 331–349.
- [4] Andrea Barbon and Angelo Rinaldo. 2024. *On The Quality Of Cryptocurrency Markets: Centralized Versus Decentralized Exchanges*. Swiss Finance Institute Research Paper 22-38. University of St. Gallen, School of Finance; Swiss Finance Institute. 71 pages. doi:10.2139/ssrn.3984897 University of St. Gallen, School of Finance Research Paper Forthcoming.
- [5] Jan Arvid Berg, Robin Fritsch, Lioba Heimbach, and Roger Wattenhofer. 2023. An Empirical Study of Market Inefficiencies in Uniswap and SushiSwap. In *Financial Cryptography and Data Security. FC 2022 International Workshops*. Springer International Publishing, Cham, 238–249.
- [6] Maxim Bichuch and Zachary Feinstein. 2024. *DeFi Arbitrage in Hedged Liquidity Tokens*. Papers 2409.11339. arXiv.org. <https://ideas.repec.org/p/arx/papers/2409.11339.html>
- [7] F. Black and M. Scholes. 1973. The Pricing of Options and Corporate Liabilities. *Journal of Political Economy* 81, 3 (1973), 637–654.
- [8] Nassib Boueri. 2021. *G3M Impermanent Loss Dynamics*. Papers 2108.06593. arXiv.org. <https://ideas.repec.org/p/arx/papers/2108.06593.html>
- [9] Vitalik Buterin. 2013. Ethereum white paper. *GitHub repository* 1, 22-23 (2013), 5–7.

- [10] Vitalik Buterin. 2014. A next-generation smart contract and decentralized application platform. *white paper* 3, 37 (14 1 2014), 2–1. [https://cryptorating.eu/Ethereum white paper](https://cryptorating.eu/Ethereum%20white%20paper).
- [11] Vitalik Buterin. 2017. Let's run on-chain decentralized exchanges the way we run prediction markets. https://www.reddit.com/r/ethereum/comments/55m04x/lets_run_onchain_decentralized_exchanges_the_way/?rdt=62847 Accessed: 2025-02-01.
- [12] Vitalik Buterin. 2021. Proposer/block builder separation-friendly fee market designs. <https://ethresear.ch/t/proposer-block-builder-separation-friendly-fee-market-designs/9725>. Accessed: 2025-10-01.
- [13] Andrea Canidio and Robin Fritsch. 2023. Arbitrageurs' profits, LVR, and sandwich attacks: batch trading as an AMM design response. arXiv:2307.02074 [q-fin.TR] Revised Feb 2025, available at <https://arxiv.org/abs/2307.02074>.
- [14] David Cao, Leonid Kogan, Gerry Tsoukalas, and Brett Hemenway Falk. 2023. *A Structural Model of Automated Market Making*. Working Paper. CBER CtCe. doi:10.2139/ssrn.4591447 Available at SSRN.
- [15] Agostino Capponi and Ruizhe Jia. 2021. The Adoption of Blockchain-based Decentralized Exchanges. arXiv:2103.08842 [q-fin.TR] <https://arxiv.org/abs/2103.08842>
- [16] Álvaro Cartea, Fayçal Drissi, and Marcello Monga. 2022. Decentralised Finance and Automated Market Making: Predictable Loss and Optimal Liquidity Provision. (November 2022). ssrn:4273989 doi:10.2139/ssrn.4273989 Forthcoming in SIAM Journal on Financial Mathematics.
- [17] Álvaro Cartea, Fayçal Drissi, and Marcello Monga. 2023. Predictable Losses of Liquidity Provision in Constant Function Markets and Concentrated Liquidity Markets. *Applied Mathematical Finance* 30, 2 (2023), 69–93. doi:10.1080/1350486X.2023.2277957
- [18] Philip Daian, Steven Goldfeder, Tyler Kell, Yunqi Li, Xueyuan Zhao, Iddo Bentov, Lorenz Breidenbach, and Ari Juels. 2019. Flash Boys 2.0: Frontrunning, Transaction Reordering, and Consensus Instability in Decentralized Exchanges. arXiv:1904.05234 [cs.CR] <https://arxiv.org/abs/1904.05234>
- [19] Freddy Delbaen and Walter Schachermayer. 2006. *The Mathematics of Arbitrage*. Springer. <https://link.springer.com/book/10.1007/3-540-33589-1>
- [20] Alex Evans, Guillermo Angeris, and Tarun Chitra. 2021. Optimal Fees for Geometric Mean Market Makers. In *Financial Cryptography and Data Security. FC 2021 International Workshops*, Matthew Bernhard, Andrea Bracciali, Lewis Gudgeon, Thomas Haines, Ariah Klages-Mundt, Shin'ichiro Matsuo, Daniel Perez, Massimiliano Sala, and Sam Werner (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 65–79.
- [21] Thierry Foucault. 1998. *Order Flow Composition and Trading Costs in Dynamic Limit Order Markets*. Discussion Paper 1817. CEPR. Available at SSRN: <https://ssrn.com/abstract=106250>.
- [22] Robin Fritsch and Andrea Canidio. 2024. Measuring Arbitrage Losses and Profitability of AMM Liquidity. In *Companion Proceedings of the ACM Web Conference 2024* (Singapore, Singapore) (WWW '24). Association for Computing Machinery, New York, NY, USA, 1761–1767. doi:10.1145/3589335.3651961
- [23] Pranav Garimidi, Joachim Neu, and Max Resnick. 2025. Multiple Concurrent Proposers: Why and How. arXiv:2509.23984 [cs.CR] <https://arxiv.org/abs/2509.23984>
- [24] Krzysztof Gogol, Johnnatan Messias, Deborah Miori, Claudio Tessone, and Benjamin Livshits. 2024. Cross-Rollup MEV: Non-Atomic Arbitrage Across L2 Blockchains. arXiv preprint arXiv:2406.02172 (2024).
- [25] Joel Hasbrouck, Thomas J Rivera, and Fahad Saleh. 2022. The Need for Fees at a DEX: How Increases in Fees Can Increase DEX Trading Volume. Available at SSRN (17 8 2022), 48. doi:10.2139/ssrn.4192925 Posted: 26 Aug 2022, Last revised: 13 Dec 2022.
- [26] Joel Hasbrouck, Thomas J Rivera, and Fahad Saleh. 2023. An Economic Model of a Decentralized Exchange with Concentrated Liquidity. (Aug 2023). doi:10.2139/ssrn.4529513 Available at SSRN: <https://ssrn.com/abstract=4529513>.
- [27] Xue Dong He, Chen Yang, and Yutian Zhou. 2025. Arbitrage on Decentralized Exchanges. doi:10.2139/ssrn.5347630 Available at SSRN: <https://ssrn.com/abstract=5347630> or <http://dx.doi.org/10.2139/ssrn.5347630>.
- [28] Lioba Heimbach, Vabuk Pahari, and Eric Schertenleib. 2024. Non-atomic arbitrage in decentralized finance. In *IEEE Symposium on Security and Privacy (SP)*. IEEE, 3866–3884.
- [29] J. Milionis, C. C. Moallemi and T. Roughgarden. 2024. A Myersonian Framework for Optimal Liquidity Provision in Automated Market Makers. In *15th Innovations in Theoretical Computer Science Conference (ITCS 2024) (LIPIcs, Vol. 287)*. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 81:1–81:19.
- [30] Bence Ládóckzi. 2024. Mean-reverting Prices and Losses in Liquidity Pools. In *2024 6th International Conference on Blockchain Computing and Applications (BCCA)*. 149–155. doi:10.1109/BCCA62388.2024.10844419
- [31] Alan Lu and Martin Köppelmann. 2017. Building a Decentralized Exchange in Ethereum. <https://blog.gnosis.pm/building-a-decentralized-exchange-in-ethereum-eea4e7452d6e>. Accessed: 2022-07-31.
- [32] R. C. Merton. 1976. Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics* 3, 1-2 (1976), 125–144.
- [33] J. Milionis, C. C. Moallemi, and T. Roughgarden. 2023. Automated market making and arbitrage profits in the presence of fees. arXiv:2305.14604
- [34] Jason Milionis, Ciamac C. Moallemi, and Tim Roughgarden. 2024. Complexity-Approximation Trade-Offs in Exchange Mechanisms: AMMs vs. LOBs. In *Financial Cryptography and Data Security*, Foteini Baldimtsi and Christian Cachin (Eds.). Springer Nature Switzerland, Cham, 326–343.
- [35] J. Milionis, C. C. Moallemi, T. Roughgarden, and A. L. Zhang. 2024. Automated Market Making and Loss-Versus-Rebalancing. arXiv (2024). arXiv:2208.06046
- [36] Alex Nezlobin and Martin Tassy. 2025. Loss-Versus-Rebalancing under Deterministic and Generalized block-times. arXiv preprint arXiv:2505.05113 (2025).
- [37] Andreas Park. 2021. The Conceptual Flaws of Constant Product Automated Market Making. SSRN Electronic Journal (01 2021). doi:10.2139/ssrn.3805750
- [38] Lukáš Pichl, Zheng Nan, and Taisei Kaizoji. 2020. *Time Series Analysis of Ether Cryptocurrency Prices: Efficiency, Predictability, and Arbitrage on Exchange Rates*. Springer Singapore, Singapore, 183–196. doi:10.1007/978-981-15-4498-9_10
- [39] Kaihua Qin, Liyi Zhou, and Arthur Gervais. 2021. Quantifying Blockchain Extractable Value: How dark is the forest? arXiv:2101.05511 [cs.CR] <https://arxiv.org/abs/2101.05511>
- [40] Lawrence R. Rabiner. 1989. A tutorial on hidden Markov models and selected applications in speech recognition. *Proc. IEEE* 77, 2 (1989), 257–286. doi:10.1109/5.18626
- [41] G. E. Uhlenbeck and L. S. Ornstein. 1930. On the Theory of the Brownian Motion. *Phys. Rev.* 36 (sep 1930), 823–841.
- [42] Uniswap. [n.d.]. Uniswap Info. <https://info.uniswap.org/>. Accessed on January 25, 2026.
- [43] Ye Wang, Yan Chen, Haotian Wu, Liyi Zhou, Shuiguang Deng, and Roger Wattenhofer. 2022. Cyclic Arbitrage in Decentralized Exchanges. In *Companion Proceedings of the Web Conference 2022 (WWW '22)*. Association for Computing Machinery, New York, NY, USA, 12–19. doi:10.1145/3487553.3524201
- [44] Fei Wu, Danning Sui, Thomas Thiery, and Malleh Pai. 2025. Measuring CEX-DEX Extracted Value and Searcher Profitability: The Darkest of the MEV Dark Forest. In *Advances in Financial Technologies (AFT)*. arXiv preprint arXiv:2507.13023.
- [45] Jiahua Xu, Krzysztof Paruch, Simon Cousaert, and Yebo Feng. 2023. SoK: Decentralized Exchanges (DEX) with Automated Market Maker (AMM) Protocols. *ACM Comput. Surv.* 55, 11, Article 238 (Feb. 2023), 50 pages. doi:10.1145/3570639
- [46] Yu Zhang, Tao Yan, Jianhong Lin, Benjamin Kraner, and Claudio J. Tessone. 2024. An Improved Algorithm to Identify More Arbitrage Opportunities on Decentralized Exchanges. In *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*. 1–7. doi:10.1109/ICBC59979.2024.10634351

A Additional Empirical Data

Date	Swaps	ARB	ARB%	Noise	ETH Vol	ARB Vol (\$10 ³)	Noise Vol (\$10 ³)	Fees (\$)	ARB Profit (\$)	Vol/Profit	μ (%)	σ (%)
09-18	313	80	1.1	233	3.1	6.2	8.2	43.2	6.9	2083.4	1.45	1.81
09-19	229	65	0.9	164	1.7	3.0	4.5	22.5	4.0	1855.8	-2.72	1.77
09-20	174	31	0.4	143	0.7	1.2	2.2	10.0	1.5	2146.9	0.42	0.95
09-21	185	31	0.4	154	0.6	0.8	2.0	8.6	0.5	5451.9	-0.42	0.98
09-22	324	129	1.8	195	6.6	15.2	12.7	83.5	89.7	310.3	-6.29	3.61
09-23	263	99	1.4	164	2.8	5.0	6.7	35.1	5.4	2161.0	-0.38	1.61
09-24	235	78	1.1	157	2.9	7.0	5.0	36.2	8.7	1392.8	-0.90	2.04
09-25	382	189	2.6	193	9.4	22.1	15.5	112.8	46.6	807.2	-5.99	4.01
09-26	331	134	1.9	197	6.7	15.6	11.0	79.7	29.3	906.2	3.01	3.12
09-27	229	42	0.6	187	1.9	2.2	5.6	23.5	4.0	1931.9	-0.23	1.42
Average	266.5	87.8	1.2	178.7	3.7	7.8	7.3	45.5	19.7	1904.8	-1.21	2.13

Table 7: Daily arbitrage analysis for 0x17C1Ae82D99379240059940093762c5e4539aba5 with GBM parameters using USD/ETH exchange rates from Binance (2025-09-18 to 2025-09-27). Shows swap activity, arbitrage opportunities, volumes (in thousand dollars), fees, profit, volume to profit ratio, daily drift (μ) and volatility (σ) in percentage. TVL $\approx$ 86 ETH.

Date	Swaps	ARB	ARB%	Noise	ETH Vol	ARB Vol (\$10 ⁴)	Noise Vol (\$10 ⁴)	Fees (\$)	ARB Profit (\$)	Vol/Profit	μ (%)	σ (%)
09-18	418	82	1.1	336	20.0	4.5	4.7	276.8	149.6	616.7	1.45	1.81
09-19	399	60	0.8	339	14.4	2.5	4.1	195.9	42.1	1551.9	-2.72	1.77
09-20	364	51	0.7	313	2.8	0.6	0.7	37.8	4.8	2653.7	0.42	0.95
09-21	321	61	0.8	260	2.7	0.6	0.7	36.7	3.6	3439.2	-0.42	0.98
09-22	642	183	2.5	459	47.1	8.7	11.2	595.6	263.5	753.3	-6.29	3.61
09-23	478	131	1.8	347	18.2	2.3	5.3	228.2	26.5	2868.8	-0.38	1.61
09-24	414	107	1.5	307	14.7	3.6	2.5	184.1	36.0	1705.7	-0.90	2.04
09-25	594	197	2.7	397	41.3	10.3	6.1	492.0	124.5	1317.5	-5.99	4.01
09-26	662	158	2.2	504	32.5	6.5	6.3	384.1	84.7	1511.7	3.01	3.12
09-27	468	38	0.5	430	5.7	0.9	1.4	68.3	12.2	1867.8	-0.23	1.42
Average	476.0	106.8	1.5	369.2	19.9	4.0	4.3	250.0	74.7	1828.6	-1.21	2.13

Table 8: Daily arbitrage analysis for 0x06da0fd433C1A5d7a4faa0111c044910A184553 with GBM parameters using USD/ETH exchange rates from Binance (2025-09-18 to 2025-09-27). Shows swap activity, arbitrage opportunities, volumes (in ten thousand dollars), fees, profit, volume to profit ratio, daily drift (μ) and volatility (σ) in percentage. TVL $\approx$ 502 ETH.

Date	Swaps	ARB	ARB%	Noise	ETH Vol	ARB Vol (\$10 ³)	Noise Vol (\$10 ³)	Fees (\$)	ARB Profit (\$)	Vol/Profit	μ (%)	σ (%)
09-18	138	61	0.8	77	0.4	0.9	0.7	5.0	1.2	1389.4	1.45	1.81
09-19	117	39	0.5	78	0.3	0.7	0.7	4.0	0.6	2126.9	-2.72	1.77
09-20	80	12	0.2	68	0.1	0.1	0.3	1.2	0.1	3135.4	0.42	0.95
09-21	92	16	0.2	76	0.1	0.1	0.4	1.4	0.1	5354.8	-0.42	0.98
09-22	218	89	1.2	129	3.3	6.5	7.3	41.5	99.4	139.2	-6.29	3.61
09-23	184	62	0.9	122	0.6	0.9	1.7	8.0	1.4	1965.8	-0.38	1.61
09-24	122	46	0.6	76	0.4	0.8	1.0	5.4	1.2	1466.4	-0.90	2.04
09-25	241	107	1.5	134	1.3	2.8	2.5	15.9	5.2	1024.6	-5.99	4.01
09-26	249	114	1.6	135	1.2	2.1	2.6	14.1	3.5	1363.4	3.01	3.12
09-27	90	17	0.2	73	0.3	0.3	0.9	3.7	0.4	3208.1	-0.23	1.42
Average	153.1	56.3	0.8	96.8	0.8	1.5	1.8	10.0	11.3	2117.4	-1.21	2.13

Table 9: Daily arbitrage analysis for 0x74C99F3f5331676f6AEc2756e1F39b4FC029a83E with GBM parameters using USD/ETH exchange rates from Binance (2025-09-18 to 2025-09-27). Shows swap activity, arbitrage opportunities, volumes (in thousand dollars), fees, profit, volume to profit ratio, daily drift (μ) and volatility (σ) in percentage. TVL $\approx$ 18 ETH.

Date	Swaps	ARB	ARB%	Noise	ETH Vol	ARB Vol (\$10 ³)	Noise Vol (\$10 ³)	Fees (\$)	ARB Profit (\$)	Vol/Profit	μ (%)	σ (%)
09-18	66	31	0.4	35	0.4	0.9	0.8	5.2	1.5	1190.8	1.45	1.81
09-19	55	27	0.4	28	1.0	0.6	3.8	13.1	2.0	2212.5	-2.72	1.77
09-20	48	19	0.3	29	0.2	0.4	0.5	2.8	0.7	1341.8	0.42	0.95
09-21	78	22	0.3	56	0.7	0.8	2.2	9.0	2.2	1350.9	-0.42	0.98
09-22	85	38	0.5	47	1.3	3.0	2.6	16.7	15.3	364.2	-6.29	3.61
09-23	54	21	0.3	33	0.3	0.7	0.7	4.3	1.1	1331.6	-0.38	1.61
09-24	68	26	0.4	42	0.6	1.3	1.3	7.8	4.8	545.8	-0.90	2.04
09-25	126	61	0.8	65	1.8	3.3	3.8	21.1	11.1	633.7	-5.99	4.01
09-26	109	60	0.8	49	1.0	2.1	1.9	12.0	8.1	491.5	3.01	3.12
09-27	44	21	0.3	23	0.7	1.4	1.5	8.7	11.4	254.2	-0.23	1.42
Average	73.3	32.6	0.5	40.7	0.8	1.4	1.9	10.1	5.8	971.7	-1.21	2.13

Table 10: Daily arbitrage analysis for 0x703b120F15Ab77B986a24c6f9262364d02f9432f with GBM parameters using USDT/ETH exchange rates from Binance (2025-09-18 to 2025-09-27). Shows swap activity, arbitrage opportunities, volumes (in thousand dollars), fees, profit, volume to profit ratio, daily drift (μ) and volatility (σ) in percentage. TVL $\approx$ 16 ETH.

Date	Swaps	ARB	ARB%	Noise	ETH Vol	ARB Vol (\$10 ³)	Noise Vol (\$10 ³)	Fees (\$)	ARB Profit (\$)	Vol/Profit	μ (%)	σ (%)
09-18	32	25	0.3	7	1.5	5.6	1.5	21.3	10.1	700.8	1.45	1.81
09-19	30	24	0.3	6	1.4	4.3	1.8	18.3	6.3	968.2	-2.72	1.77
09-20	12	8	0.1	4	0.4	1.1	0.9	6.0	1.3	1571.2	0.42	0.95
09-21	20	11	0.2	9	0.5	1.0	1.2	6.6	0.8	2675.5	-0.42	0.98
09-22	77	53	0.7	24	5.3	15.6	6.9	67.4	70.6	318.3	-6.29	3.61
09-23	58	37	0.5	21	2.3	6.2	3.4	29.0	7.2	1332.4	-0.38	1.61
09-24	39	28	0.4	11	1.9	6.3	1.8	24.3	8.3	971.3	-0.90	2.04
09-25	111	82	1.1	29	7.1	20.7	7.5	84.6	47.8	590.4	-5.99	4.01
09-26	78	59	0.8	19	4.6	12.5	5.5	53.9	21.7	827.2	3.01	3.12
09-27	19	14	0.2	5	0.9	2.7	1.0	11.0	3.6	1020.7	-0.23	1.42
Average	47.6	34.1	0.5	13.5	2.6	7.6	3.2	32.2	17.8	1097.6	-1.21	2.13

Table 11: Daily arbitrage analysis for 0x3Af7A58D54cf014675af2B7EBc222C3166BF5692 with GBM parameters using USDT/ETH exchange rates from Binance (2025-09-18 to 2025-09-27). Shows swap activity, arbitrage opportunities, volumes (in thousand dollars), fees, profit, volume to profit ratio, daily drift (μ) and volatility (σ) in percentage. TVL $\approx$ 109 ETH.

Date	Swaps	ARB	ARB%	Noise	ETH Vol	ARB Vol (\$10 ⁴)	Noise Vol (\$10 ⁴)	Fees (\$)	ARB Profit (\$)	Vol/Profit	μ (%)	σ (%)
09-18	26	25	0.3	1	2.1	0.9	0.0	28.3	12.9	729.9	1.45	1.81
09-19	19	18	0.2	1	1.7	0.7	0.0	22.5	8.9	836.9	-2.72	1.77
09-20	6	5	0.1	1	0.3	0.2	0.0	4.7	3.2	488.3	0.42	0.95
09-21	14	7	0.1	7	0.5	0.1	0.1	6.5	1.0	2278.7	-0.42	0.98
09-22	59	48	0.7	11	6.7	2.5	0.4	86.1	104.2	275.3	-6.29	3.61
09-23	41	28	0.4	13	2.6	0.8	0.3	32.4	9.7	1119.6	-0.38	1.61
09-24	35	32	0.4	3	2.8	1.1	0.1	34.4	13.6	845.4	-0.90	2.04
09-25	60	52	0.7	8	7.8	2.7	0.4	93.7	60.3	517.8	-5.99	4.01
09-26	65	54	0.8	11	6.5	2.3	0.3	76.6	43.9	581.1	3.01	3.12
09-27	20	13	0.2	7	1.3	0.4	0.1	15.3	5.8	873.7	-0.23	1.42
Average	34.5	28.2	0.4	6.3	3.2	1.2	0.2	40.1	26.4	854.7	-1.21	2.13

Table 12: Daily arbitrage analysis for 0x008D9d457AD0A75c83D9a5e2b8E4Ce89232b5083 with GBM parameters using USDT/ETH exchange rates from Binance (2025-09-18 to 2025-09-27). Shows swap activity, arbitrage opportunities, volumes (in ten thousand dollars), fees, profit, volume to profit ratio, daily drift (μ) and volatility (σ) in percentage. TVL $\approx$ 161 ETH.

Date	Swaps	ARB	ARB%	Noise	ETH Vol	ARB Vol (\$10 ³)	Noise Vol (\$10 ³)	Fees (\$)	ARB Profit (\$)	Vol/Profit	μ (%)	σ (%)
09-18	13	13	0.2	0	0.2	1.0	0.0	3.1	2.8	367.2	1.45	1.81
09-19	10	9	0.1	1	0.2	0.9	0.0	2.7	1.7	510.5	-2.72	1.77
09-20	2	2	0.0	0	0.0	0.1	0.0	0.4	0.2	507.9	0.42	0.95
09-21	2	1	0.0	1	0.0	0.1	0.0	0.2	0.1	1141.2	-0.42	0.98
09-22	29	24	0.3	5	1.3	3.0	2.4	16.3	18.4	295.4	-6.29	3.61
09-23	11	9	0.1	2	0.2	0.5	0.1	2.0	1.1	631.1	-0.38	1.61
09-24	14	13	0.2	1	0.2	0.9	0.1	3.1	2.0	518.2	-0.90	2.04
09-25	20	20	0.3	0	0.6	2.3	0.0	6.8	8.1	282.4	-5.99	4.01
09-26	19	19	0.3	0	0.6	2.3	0.0	6.9	8.1	283.4	3.01	3.12
09-27	4	2	0.0	2	0.1	0.2	0.1	0.7	0.3	808.7	-0.23	1.42
Average	12.4	11.2	0.2	1.2	0.3	1.1	0.3	4.2	4.3	534.6	-1.21	2.13

Table 13: Daily arbitrage analysis for 0x504393f35e3cF04977845E183841D8e4Ec0eF6b3 with GBM parameters using USDT/ETH exchange rates from Binance (2025-09-18 to 2025-09-27). Shows swap activity, arbitrage opportunities, volumes (in thousand dollars), fees, profit, volume to profit ratio, daily drift (μ) and volatility (σ) in percentage. TVL $\approx$ 19.4 ETH.